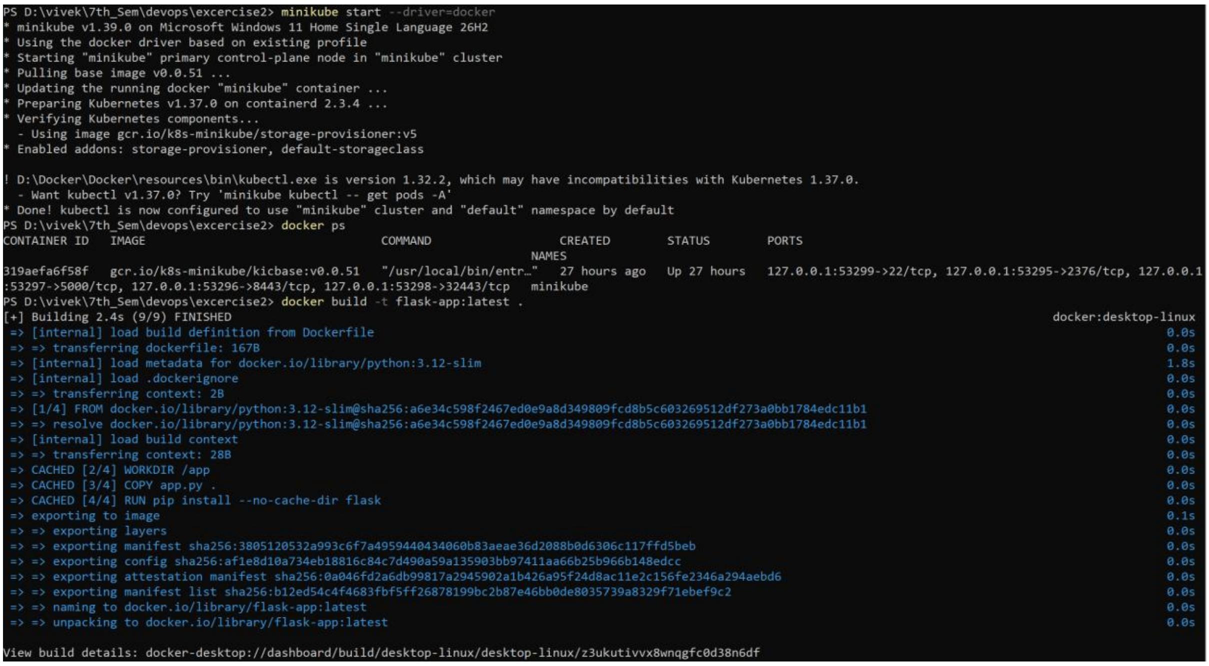
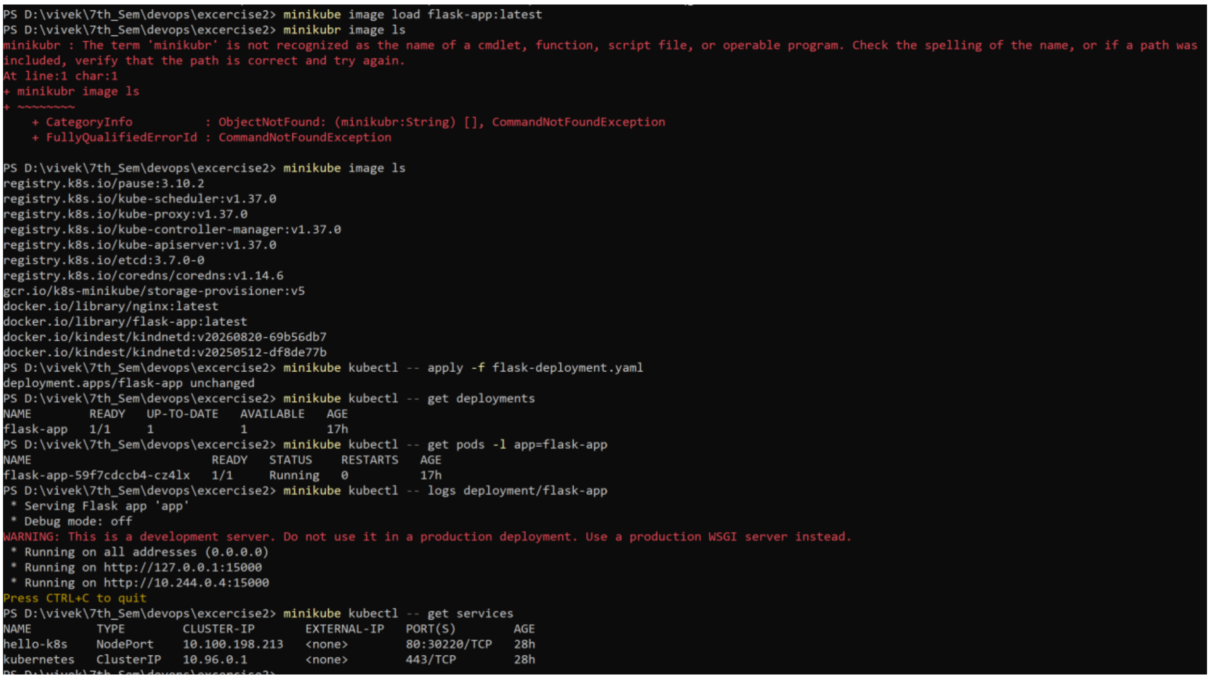
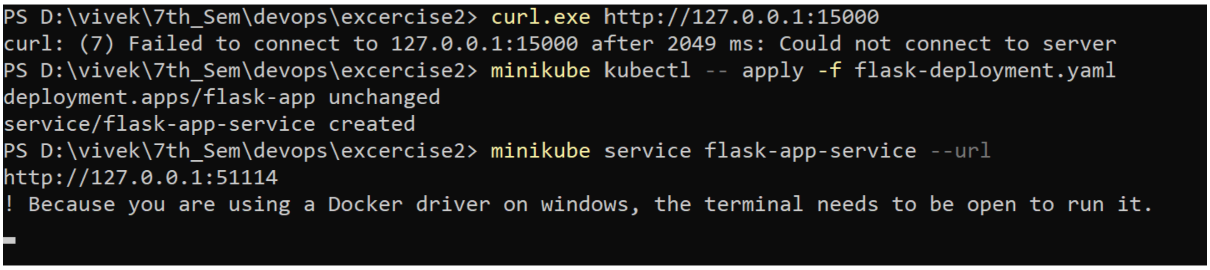
Devops – Exercise 2



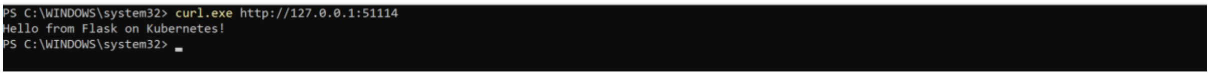
```
PS D:\vivek\7th_Sem\devops\excercise2> minikube start --driver=docker
* minikube v1.39.0 on Microsoft Windows 11 Home Single Language 26H2
* Using the docker driver based on existing profile
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.51 ...
* Updating the running docker "minikube" container ...
* Preparing Kubernetes v1.37.0 on containerd 2.3.4 ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: storage-provisioner, default-storageclass

! D:\Docker\Docker\resources\bin\kubectl.exe is version 1.32.2, which may have incompatibilities with Kubernetes 1.37.0.
  - Want kubectl v1.37.0? Try 'minikube kubectl -- get pods -A'
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
PS D:\vivek\7th_Sem\devops\excercise2> docker ps
CONTAINER ID   IMAGE                                 COMMAND                  CREATED        STATUS        PORTS
                                                                         NAMES
319aefa6f58f   gcr.io/k8s-minikube/kicbase:v0.0.51   "/usr/local/bin/entr…"   27 hours ago   Up 27 hours   127.0.0.1:53299->22/tcp, 127.0.0.1:53295->2376/tcp, 127.0.0.1
:53297->5000/tcp, 127.0.0.1:53296->8443/tcp, 127.0.0.1:53298->32443/tcp   minikube
PS D:\vivek\7th_Sem\devops\excercise2> docker build -t flask-app:latest .
[+] Building 2.4s (9/9) FINISHED                                                                                              docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                                           0.0s
 => => transferring dockerfile: 167B                                                                                                           0.0s
 => [internal] load metadata for docker.io/library/python:3.12-slim                                                                            1.8s
 => [internal] load .dockerignore                                                                                                              0.0s
 => => transferring context: 2B                                                                                                                0.0s
 => [1/4] FROM docker.io/library/python:3.12-slim@sha256:a6e34c598f2467ed0e9a8d349809fcd8b5c603269512df273a0bb1784edc11b1                      0.0s
 => => resolve docker.io/library/python:3.12-slim@sha256:a6e34c598f2467ed0e9a8d349809fcd8b5c603269512df273a0bb1784edc11b1                      0.0s
 => [internal] load build context                                                                                                              0.0s
 => => transferring context: 28B                                                                                                               0.0s
 => CACHED [2/4] WORKDIR /app                                                                                                                  0.0s
 => CACHED [3/4] COPY app.py .                                                                                                                 0.0s
 => CACHED [4/4] RUN pip install --no-cache-dir flask                                                                                          0.0s
 => exporting to image                                                                                                                         0.1s
 => => exporting layers                                                                                                                        0.0s
 => => exporting manifest sha256:3805120532a993c6f7a4959440434060b83aeae36d2088b0d6306c117ffd5beb                                              0.0s
 => => exporting config sha256:af1e8d10a734eb18816c84c7d490a59a135903bb97411aa66b25b966b148edcc                                                0.0s
 => => exporting attestation manifest sha256:0a046fd2a6db99817a2945902a1b426a95f24d8ac11e2c156fe2346a294aebd6                                  0.0s
 => => exporting manifest list sha256:b12ed54c4f4683fbf5ff26878199bc2b87e46bb0de8035739a8329f71ebef9c2                                         0.0s
 => => naming to docker.io/library/flask-app:latest                                                                                            0.0s
 => => unpacking to docker.io/library/flask-app:latest                                                                                         0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/z3ukutivvx8wnqgfc0d38n6df
```



```
PS D:\vivek\7th_Sem\devops\excercise2> minikube image load flask-app:latest
PS D:\vivek\7th_Sem\devops\excercise2> minikubr image ls
minikubr : The term 'minikubr' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was
included, verify that the path is correct and try again.
At line:1 char:1
+ minikubr image ls
+ ~~~~~~~~
    + CategoryInfo          : ObjectNotFound: (minikubr:String) [], CommandNotFoundException
    + FullyQualifiedErrorId : CommandNotFoundException

PS D:\vivek\7th_Sem\devops\excercise2> minikube image ls
registry.k8s.io/pause:3.10.2
registry.k8s.io/kube-scheduler:v1.37.0
registry.k8s.io/kube-proxy:v1.37.0
registry.k8s.io/kube-controller-manager:v1.37.0
registry.k8s.io/kube-apiserver:v1.37.0
registry.k8s.io/etcd:3.7.0-0
registry.k8s.io/coredns/coredns:v1.14.6
gcr.io/k8s-minikube/storage-provisioner:v5
docker.io/library/nginx:latest
docker.io/library/flask-app:latest
docker.io/kindest/kindnetd:v20260820-69b56db7
docker.io/kindest/kindnetd:v20250512-df8de77b
PS D:\vivek\7th_Sem\devops\excercise2> minikube kubectl -- apply -f flask-deployment.yaml
deployment.apps/flask-app unchanged
PS D:\vivek\7th_Sem\devops\excercise2> minikube kubectl -- get deployments
NAME        READY   UP-TO-DATE   AVAILABLE   AGE
flask-app   1/1     1            1           17h
PS D:\vivek\7th_Sem\devops\excercise2> minikube kubectl -- get pods -l app=flask-app
NAME                         READY   STATUS    RESTARTS   AGE
flask-app-59f7cdccb4-cz4lx   1/1     Running   0          17h
PS D:\vivek\7th_Sem\devops\excercise2> minikube kubectl -- logs deployment/flask-app
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:15000
 * Running on http://10.244.0.4:15000
Press CTRL+C to quit
PS D:\vivek\7th_Sem\devops\excercise2> minikube kubectl -- get services
NAME         TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
hello-k8s    NodePort    10.100.198.213   <none>        80:30220/TCP   28h
kubernetes   ClusterIP   10.96.0.1        <none>        443/TCP        28h
PS D:\vivek\7th_Sem\devops\excercise2>
```



```
PS D:\vivek\7th_Sem\devops\excercise2> curl.exe http://127.0.0.1:15000
curl: (7) Failed to connect to 127.0.0.1:15000 after 2049 ms: Could not connect to server
PS D:\vivek\7th_Sem\devops\excercise2> minikube kubectl -- apply -f flask-deployment.yaml
deployment.apps/flask-app unchanged
service/flask-app-service created
PS D:\vivek\7th_Sem\devops\excercise2> minikube service flask-app-service --url
http://127.0.0.1:51114
! Because you are using a Docker driver on windows, the terminal needs to be open to run it.
```



```
PS C:\WINDOWS\system32> curl.exe http://127.0.0.1:51114
Hello from Flask on Kubernetes!
PS C:\WINDOWS\system32>
```